

Instinct and learning

When turtles hatch, they scuttle straight toward the sea. Newly hatched birds beg for food from their parents, even though they cannot see. Both are examples of animal instinct—a kind of behavior that is inherited and that does not have to be learned. Instinct can make simple creatures behave in amazingly complex ways but, in most animals, instinct works hand in hand with learning. Knowledge and experience make animals better at the things they do and equip them with the skills needed for survival. Learning lets some animals develop remarkable new abilities, including making and using tools.

Feeding trigger

American robins are helpless when they hatch. Instinct makes the chicks stay still and quiet while their parents are away from the nest. The moment a parent returns with a worm, however, the young open their beaks wide and noisily beg for food. The parent bird's feeding instincts are then triggered in turn. It places the worm into the nearest open beak, guided by the brightly colored lining of the nestling's mouth and throat.



Instinct at work

Instinctive behavior is often made up of short, simple routines. Added together, they enable animals to perform astonishing feats. However, unlike learned behavior, instinctive routines are mostly fixed. For example, a potter wasp always builds the same kind of potlike nest. It does not plan ahead while it is working, and it never invents a new method.



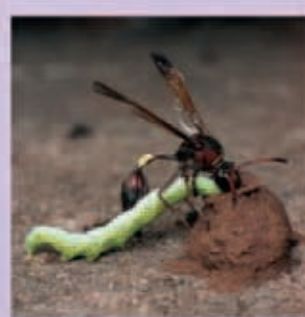
Moving on

Common cuckoos migrate long distances each year between Europe and Africa, guided by instinct. Even on their first journey, they travel alone.



Hiding away

Young roe deer spend their first few weeks hidden from predators in long grass. Instinct makes them keep completely still, even if danger threatens.



Feeding young

Female potter wasps lay their eggs in clay nurseries. They stock each pot with caterpillars to feed the grub when it hatches. Then, they seal up the door.

Headlong rush

After digging their way out of their nest in the sand, young loggerhead turtles move quickly to avoid being caught by predators. The hatchlings usually emerge on the beach after dark. Their instincts guide them toward the lightest part of the horizon and the sea. Unfortunately, many turtles are easily lured off course by the lights of beachside hotels and bars.



Smart cookies

Chimpanzees are our closest living relatives. They have high intelligence and use it to make different tools. Here, one chimp has made a digging stick and is using it to remove termites from a hole in their nest.

The other chimp is watching closely—a sign that it will copy the technique itself when it wants to find food. As humans, we take this kind of learning for granted, but such behavior is much rarer in the animal world.

